

Sergei Druzhkov

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EDUCATION

- **SPbSU (Department of Mathematics and Computer Sciences)** Saint Petersburg, Russia
Applied Mathematics and Computer Science (Modern Programming) 2021 – 2025
GPA: 4.93 (out of 5.00)
Thesis: Clone detection and automatic refactoring of Lua programs for network devices
- **ITMO (Institute of Applied Computer Sciences)** Saint Petersburg, Russia
Applied Mathematics and Computer Science (Programming and Artificial Intelligence) 2025 – 2027
GPA: 5.00 (out of 5.00)
Thesis: —

SKILLS SUMMARY

- **Languages:** C/C++, SQL, Go, TypeScript, Lua, Python
- **Technologies:** LLVM, SQLite, CMake, Make, Git, Docker, $\LaTeX$
- **Communication:** Russian (Native), English (Upper-Intermediate)

EXPERIENCE

- **Huawei Russian Research Institute**
Research engineer April 2022 – Present
 - **Engineering activity:** Development of high-performance applications based on research and classical results, as well as their support and integration into the company's existing ecosystem.
 - **Research activity:** Analysis of academic publications and conducting own research based on them in the field of program analysis and developer tools.

PROJECTS

- **Static analysis tool for C/C++:**
 - Interprocedural taint analysis of C/C++ programs based on LLVM/Clang for large codebases (>20 MLOC).
 - Development of domain-specific checks and solutions for internal company projects.
 - Research on applicability of graph databases (ArangoDB, Neo4j) for program analysis tasks.
- **Lua minifier:**
 - Classic code reduction techniques: removing whitespace, comments, and blank lines; renaming variables and labels; dead code elimination and other language-specific code transformations.
 - Ability to find and remove duplicate code fragments.
 - Code size reduction by more than 2 times. Runtime under 2 minutes on a large codebase (5 MLOC).
- **Debugging tools for C/C++:**
 - Support for debugging C/C++ code with GDB/LLDB on Linux/Windows/HarmonyOS.
 - Remote device debugging via SSH and Telnet.
 - Advanced debugging techniques (child process debugging, deadlock detector, thread fuzzer, memory debugging, etc.).
 - Over 1500 regular users inside the company.
- **Debug Agent:**
 - MCP server based on VS Code API for interacting with the debugging subsystem with full integration into VS Code UI.
 - MCP server for CLI agents to work with debuggers such as GDB, LLDB, Delve, etc. via DAP.

HONORS AND AWARDS

- Participant of the All-Russian Olympiad in Mathematics — 2019 (71/130)
- Prize-winner of the All-Russian Olympiad in Mathematics — 2021 (44/166)
- Prize-winner of the All-Russian Olympiad in Economics — 2019 (13/77), 2021 (43/135)
- Prize-winner of the Regional Mathematics Olympiad for university students of Saint Petersburg — 2022 (12/122)
- Prize-winner of “I am a Professional” in “Mathematics” — 2025
- Winner of the Siberian Mathematical Olympiad — 2025 (8/378)
- Prize-winner of North Countries Universities Mathematical Competition — 2026 (24/149)

OTHER EXPERIENCE

- Teacher at Kirov Summer Multidisciplinary School (LMSH) — 2021
- Jury of the municipal and regional stage of the All-Russian Olympiad in Mathematics (Kirov) — 2021–2025
- Jury of the Saint Petersburg Mathematical Olympiad — 2022–2026